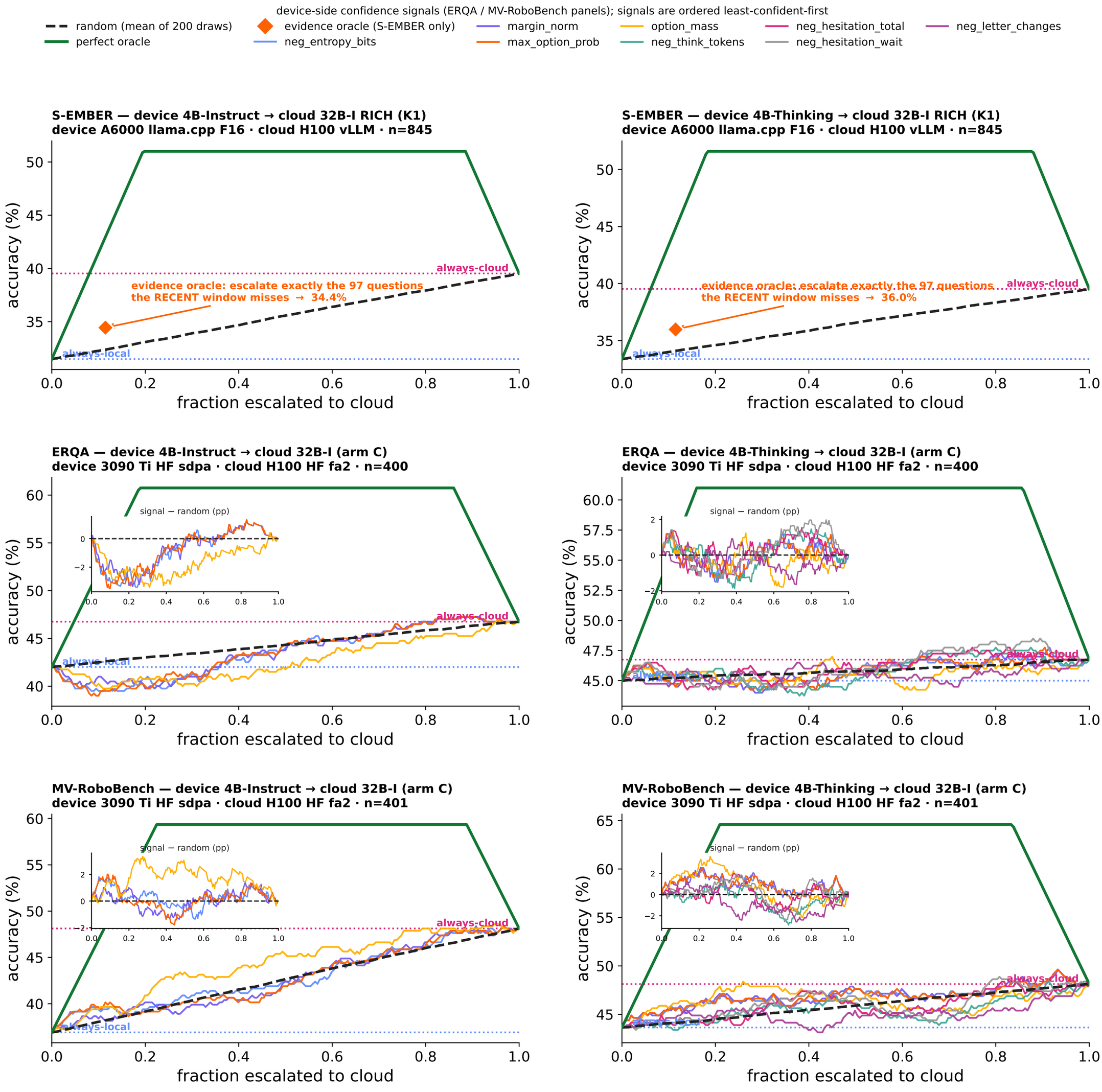
Can we predict when to escalate? Accuracy vs fraction escalated from device (Orin-class 4B) to cloud (H100 32B)

ERQA and MV-RoboBench cloud tier is 32B-Instruct (arm C), not the 32B-Thinking arm D used in reports/r4_device_signals_prep.md



AUROC of each device-side confidence signal for device correctness (Mann-Whitney U, tie-corrected; 2,000-resample question bootstrap, seed 0)

signal (higher = more confident)	AUROC for device correctness	95% CI	oracle crossing
ERQA / 4B-Instruct (n=400)			
neg_entropy_bits	0.5734	[0.5140, 0.6306]	
max_option_prob	0.5709	[0.5130, 0.6323]	
margin_norm	0.5703	[0.5121, 0.6303]	
option_mass	0.5171	[0.4611, 0.5751]	
always-local / always-cloud (arm C)	42.0% / 46.8%		perfect oracle reaches always-cloud at e = 0.048
ERQA / 4B-Thinking (n=400)			
neg_hesitation_wait	0.6229	[0.5690, 0.6747]	
neg_hesitation_total	0.6155	[0.5598, 0.6701]	
neg_think_tokens	0.6002	[0.5428, 0.6566]	
max_option_prob	0.5666	[0.5135, 0.6238]	
neg_entropy_bits	0.5645	[0.5053, 0.6194]	
margin_norm	0.5644	[0.5070, 0.6220]	
neg_letter_changes	0.5640	[0.5063, 0.6201]	
option_mass	0.5020	[0.4432, 0.5581]	
always-local / always-cloud (arm C)	45.0% / 46.8%		perfect oracle reaches always-cloud at e = 0.018
MV-RoboBench / 4B-Instruct (n=401)			
option_mass	0.6145	[0.5569, 0.6675]	
neg_entropy_bits	0.5913	[0.5345, 0.6483]	
max_option_prob	0.5902	[0.5315, 0.6482]	
margin_norm	0.5885	[0.5307, 0.6467]	
always-local / always-cloud (arm C)	36.9% / 48.1%		perfect oracle reaches always-cloud at e = 0.112
MV-RoboBench / 4B-Thinking (n=401)			
margin_norm	0.6021	[0.5450, 0.6575]	
max_option_prob	0.6012	[0.5461, 0.6554]	
neg_entropy_bits	0.6010	[0.5474, 0.6562]	
neg_hesitation_wait	0.5814	[0.5246, 0.6369]	
neg_hesitation_total	0.5698	[0.5131, 0.6261]	
option_mass	0.5594	[0.5019, 0.6159]	
neg_think_tokens	0.5421	[0.4832, 0.5996]	
neg_letter_changes	0.5070	[0.4513, 0.5587]	
always-local / always-cloud (arm C)	43.6% / 48.1%		perfect oracle reaches always-cloud at e = 0.045

Footnote: the perfect oracle escalates exactly the questions the device gets wrong and the cloud gets right. On 4–5 option multiple choice both tiers also agree by chance, so the oracle counts chance-correct cloud answers as recoverable and overstates the headroom a real trigger could reach.